

University Administration Management System (UAMS)

1. Executive Summary

This project proposes the development of a **University Administration Management System (UAMS)** to support the digital management of academic and administrative operations within a university environment.

The system will centralize key processes including **student management, staff administration, admissions, course management, attendance, examinations and results, fees and payments, timetables, reporting, and communication** into one secure platform.

The proposed system will use **Next.js and React for the frontend, Node.js for the backend, and Supabase/PostgreSQL for data management and supporting backend services**. The project will be managed through **GitHub** and deployed using **Vercel**.

The system will be developed in phases, with institutional policies, workflows, and operational requirements confirmed before final implementation.

2. Problem Statement

University administrative processes can become fragmented across manual records, separate systems, and departmental workflows. This can result in delays, inconsistent records, limited visibility of student information, difficulties in tracking fees and payments, and inefficient reporting.

A centralized administration system can help improve **efficiency, data consistency, accessibility, and decision-making** across the institution.

3. Objectives

The main objectives of the system are to:

- Centralize student, staff, and academic information.
- Simplify admissions, enrollment, and course registration.
- Support attendance, examinations, grading, and academic records.
- Improve fee, payment, and financial record management.
- Support **M-Pesa and bank transaction processing/reconciliation**.
- Provide secure, role-based access to university information.
- Provide dashboards and reports for administrative decision-making.
- Establish a scalable foundation for future university services.

4. Proposed System Modules

4.1 Student Management

Management of student profiles, admission details, enrollment status, academic information, and student records.

4.2 Staff and Faculty Management

Management of staff and lecturer records, departmental assignments, responsibilities, and system access.

4.3 Admissions and Enrollment

Support for student applications, admissions, intake management, and enrollment processes.

4.4 Course and Academic Management

Management of schools/faculties, departments, programs, courses, academic years, semesters, and course registration.

4.5 Timetable and Attendance

Management of class schedules, teaching allocations, venues, and student attendance records.

4.6 Examinations and Academic Records

Management of assessments, marks, grades, academic performance, results, and transcript generation.

4.7 Fees and Finance

Management of fee structures, student balances, invoices, payment records, and financial reconciliation.

4.8 M-Pesa and Bank Payments

Support for recording and reconciling M-Pesa and bank transactions against student financial accounts. Live integrations will depend on the institution's approved payment channels and access to required APIs.

4.9 Reporting and Dashboards

Dashboards and reports covering student records, academic performance, finances, and administrative activities.

4.10 Role-Based Access Control

Different access levels for administrators, registrar/academic staff, finance officers, lecturers, and

students to protect sensitive information.

5. Proposed Technology

Area	Technology
Frontend	Next.js, React, TypeScript
Backend	Node.js
Database & Backend Services	Supabase / PostgreSQL
Source Control	GitHub
Deployment	Vercel
Payments	M-Pesa and Bank Transactions

The architecture will allow the system to be developed as a modular application that can be expanded as additional institutional requirements are identified.

6. Security and Governance

Security will be incorporated throughout the system through:

- Authentication and role-based authorization.
- Protection of academic and financial information.
- Server-side access control for sensitive operations.
- Database security controls, including appropriate Row Level Security.
- Secure handling of system credentials and payment information.
- Audit records for important administrative and financial activities.

Institution-specific rules such as **grading methods, attendance requirements, approval workflows, and payment configurations** will remain configurable until confirmed by the university.

7. Proposed Development Approach

The system will be developed incrementally:

1. **Requirements and system design**
2. **Authentication and user management**
3. **Student and academic management**

4. **Admissions, enrollment, registration, timetable, and attendance**
5. **Examinations, results, and academic records**
6. **Fees, payments, and financial management**
7. **Testing, security review, deployment, and documentation**

This approach will allow each major component to be reviewed and improved before moving to the next stage.

8. Future Enhancements

Depending on institutional requirements, future versions may include:

- Live M-Pesa integration and automated payment reconciliation.
 - Advanced bank API integration.
 - QR-based attendance.
 - Student ID card generation.
 - Advanced management and financial reports.
 - Automated notifications and reminders.
 - Additional integrations with existing university systems.
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9. Expected Value

The proposed system is intended to:

- Reduce manual administrative work.
 - Improve accuracy and accessibility of university records.
 - Simplify academic and financial processes.
 - Improve visibility of student and institutional information.
 - Support faster reporting and decision-making.
 - Provide a scalable foundation for future digital services.
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10. Conclusion

The **University Administration Management System (UAMS)** is proposed as a practical and scalable platform for centralizing essential university academic and administrative processes.

The system will combine **academic management, student services, financial management, secure access, and reporting** within one platform while remaining flexible enough to accommodate the university's specific policies and future requirements.

This proposal is submitted for **supervisor review and feedback on the proposed scope, modules, technology, and development approach before implementation begins.**